F1 Sports Analysis: Statistical Analysis of Driver Risk Profiles and Team Reliability.

Using logistic regression, k-means and correlation to analyse team reliability and driver profiles.

Formula 1 motorsport has been around since the late 1900s, and a lot goes into making the perfect car. Each season, the teams design and engineer cars to go faster by even a fraction of a second — and that is what ultimately wins championships. But to win, both the **drivers** and the **teams** matter.

To select the best among them, we analysed **35 years of Formula One statistics** to answer three main questions on **team reliability** and **driver risk**.

Team Reliability Analysis

Main Problem Statement

What makes a Formula 1 team reliable?

Team performance changes year to year with new engines, designs, and regulations. However, over a span of 3–4 years, most teams retain the same drivers. Top teams hire the most competent drivers — often based not on age or experience, but on performance in secondary motorsport series. Hence, we created a **reliability measure** for top teams every five years, based on their starting and ending grid positions.

How It Works

Here we have analysed F1 entries across 76 seasons starting from calculating some basic statistics and performed a **Exploratory Data Analysis.** Our EDA gives us the key metrics from

1. Mean and median of starting and finishing position.
2. Mean of Qualifying position and points scored.
3. We have calculated the Dnf Rate using boolean values.
4. The grid improvement per race with results of +1.85 indicating that drivers typically gain 2 positions and +1.46 for Qualification to Race which represents the true car pace.

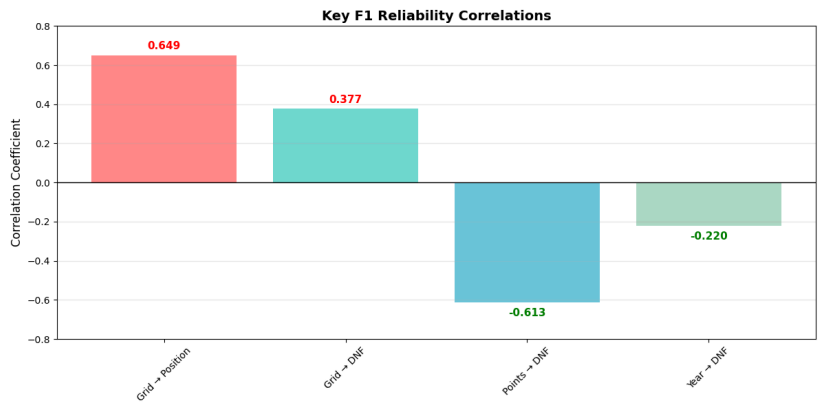
A temporal analysis becomes the fundamental key here to know the average and total number of races for teams in a season.

This EDA provides the key metrics needed for the next step — **correlation analysis**.

Correlation Analysis

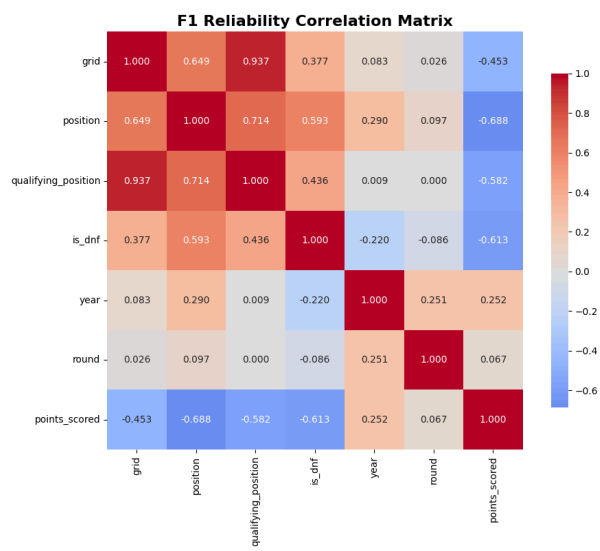
Correlation analysis helps us identify which dependent variables are influencing the team reliability. It gives us an understanding of whether performance depends more on the starting grid position, seasonal progression, or consistency over time.

We calculate the **Pearson correlation coefficient** for four key metric pairs:



These results show that starting position has a moderately strong positive correlation with finishing position, indicating that grid placement does affect the outcomes.

However this larger correlation matrix gives us an idea of all the variables the program has analysed including qualifying position, rounds, points scored and more.



Here is a code snippet for the program run:

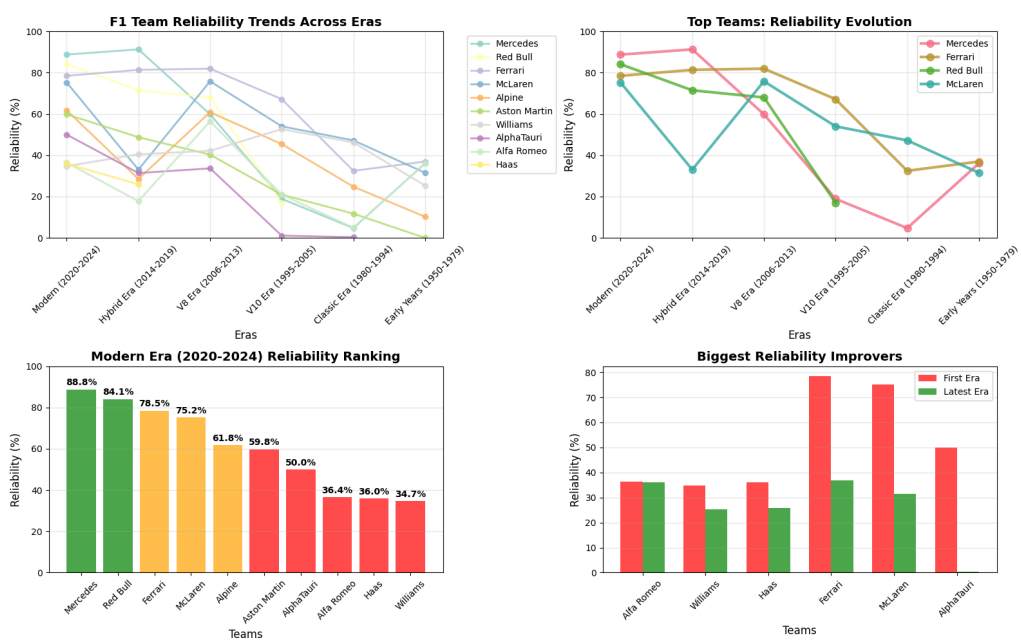
```
# === CORRELATION ANALYSIS ===
import pandas as pd

# Select numeric columns for correlation
numeric_cols = ['grid', 'position', 'qualifying_position', 'is_dnf',
'year', 'round', 'points_scored']
correlation_data = df[numeric_cols]

# Compute correlation matrix
correlation_matrix = correlation_data.corr()
print(correlation_matrix.round(3))
```

Reliability Model

Finally, we coded an input-based system that calculates average team reliability using:

Team Reliability, Average Start, Average Finish and Position Change

This helps us to evaluate each team's consistency, trends, and an overall reliability metric across multiple seasons.



Following is a code snippet for the same:

```
def analyze_team_by_eras(team_name, data):
    """Return reliability by era as a dataframe."""
    team_data = data[data['current_team'] == team_name]

    eras = {
        '2020-2024': (2020, 2024),
        '2015-2019': (2015, 2019),
        '2010-2014': (2010, 2014),
        '2000-2009': (2000, 2009),
        '1990s': (1990, 1999),
        'Pre-1990': (1950, 1989)
    }

    era_reliability = {}
    for era, (start, end) in eras.items():
        era_data = team_data[(team_data['year'] >= start) &
(team_data['year'] <= end)]
        if len(era_data) == 0:
            continue
        total = len(era_data)
        finished = len(era_data[era_data['is_dnf'] == False])
        era_reliability[era] = round(finished / total, 3)

    return pd.DataFrame([era_reliability], index=[team_name])
```

Following our analysis we can get answers to how the teams has fared, such as Ferrari has improved a lot whereas Mercedes remains the most reliable. Also teams such as Aston Martin and Haas have continued to be subpar.

```
================================================================
  ERA-BASED RELIABILITY: MERCEDES
================================================================
Last 5 Years (2020-2024)   88.8% A        214 races
                          Avg: 5.3 → 4.4 (+0.9 pos)
5 Years Before (2015-2019)  92.6% A+       204 races
                          Avg: 3.0 → 2.4 (+0.6 pos)
5 Years Before That (2010-2014)  72.4% B        192 races
                          Avg: 6.6 → 5.5 (+1.2 pos)
Early 2000s (2000-2009)    38.7% F        344 races
                          Avg: 7.7 → 5.5 (+2.2 pos)
1990s Era                   4.0% F        321 races
                          Avg: 9.9 → 5.8 (+4.1 pos)
Pre-1990                   21.9% F        680 races
                          Avg: 7.2 → 3.3 (+3.9 pos)
```

Rookie Driver Risk Clustering

Main Problem Statement

Reliable or high-scoring teams often sign rookie drivers based on their performance in a preliminary motorsport series. However more than often these high points or championship titles come with crashes, making collision or crash history a key consideration while evaluating the potential recruits.

```
 F2 CLUSTER RESULTS FOR GABRIEL BORTELETO:
========================================
CLUSTER: 2
RISK LEVEL: High Risk

Based on F2 Statistics:
 Championships: 1
 Years: 1
 Race Starts: 28
 Race Wins: 2
 Podiums: 5
 Crash in 1st Race: No
 DNF in 1st Race: No
```

How It Works

We conduct a **driver risk assessment** using a comprehensive dataset containing:
No of Years, Crash in 1st Race, DNF in 1st Race, DNS in 1st Race, Wins, Podiums, Total Races, Total Championships titles won in junior series, Total Crashes or Collisions, and Total DNF per Driver — covering 125 drivers from their preliminary racing series.

Using these 8 parameters, we use the code to do a k-means in 8 Dimensional Space and target 3 cluster groups.

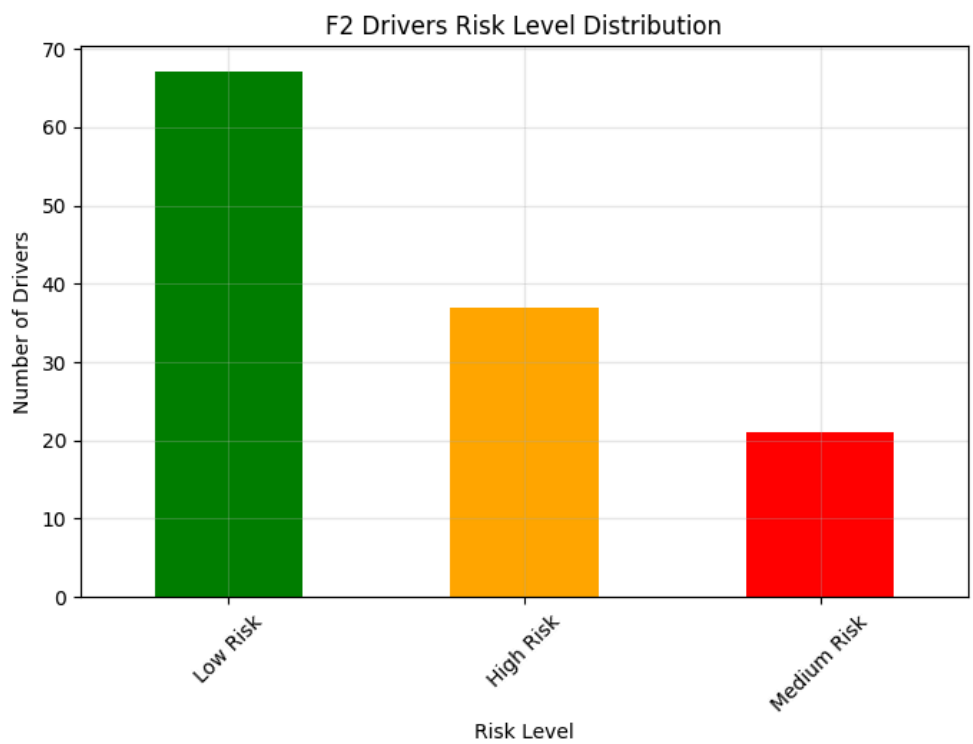
```
from sklearn.cluster import KMeans
from sklearn.preprocessing import StandardScaler

scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

# K-Means clustering in 8D space
kmeans = KMeans(n_clusters=3, random_state=42)
clusters = kmeans.fit_predict(X_scaled)
```

- Before clustering, we apply a **StandardScaler** to normalize the data so that all variables are on the same scale.
- After fitting the K-Means model, we continue to analyze cluster profiles using key metrics we have calculated preliminarily such as Crash rate, DNF rate, Win rate, and Podium

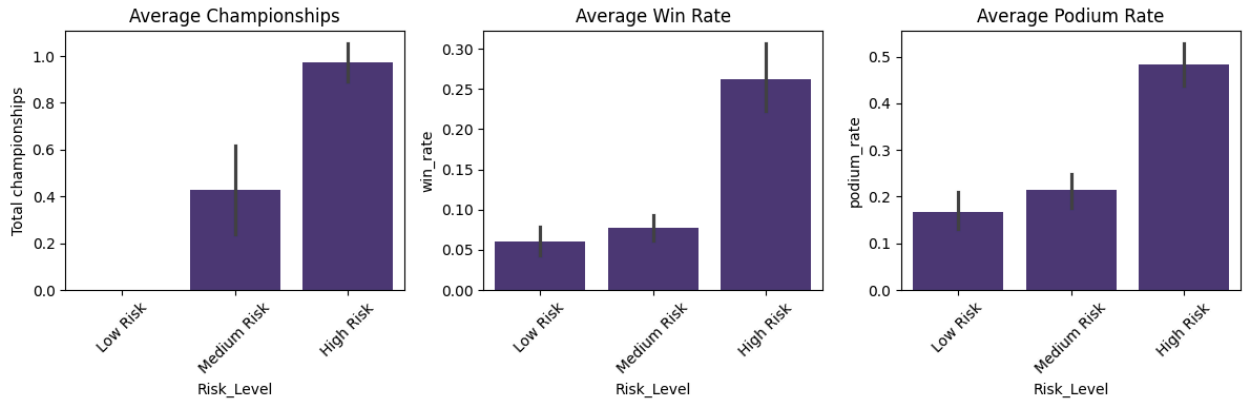
rate. The average statistics of each of these clusters help us estimate the risk parameters.

- We then go on to assign a **risk rank** to each cluster based on crash rates and perform an unsupervised cluster assignment for the new rookie drivers using their preliminary series statistics.

```
cluster_profiles = df.groupby('risk_cluster').agg({
    'crash_rate': 'mean',
    'dnf_rate': 'mean',
    'win_rate': 'mean',
    'podium_rate': 'mean'
}).round(3)

# Rank clusters: higher crash_rate → higher risk
risk_order =
cluster_profiles['crash_rate'].sort_values(ascending=False).index
```



This visualization shows how drivers are grouped into three distinct risk clusters.



Rookie Point Prediction

Main Problem Statement

Based on what has worked for other drivers, are we able to forecast if a rookie driver will score points in their debut Formula 1 season or not?

The key factors in winning the championship title is total points scored. Hence it becomes essential for newly recruited rookie drivers to score points in lieu of gaining sponsors and to have a stake in the championship title. Our goal in this analysis is to determine if rookie drivers will score points in their first Formula 1 season.

How It Works

Before training the model, we clean and standardize all driver data to ensure we have consistency across different eras and sources. Our dataset consists of 125 drivers total split into 75% or 94 drivers for fitting the model and another 25% or 31 drivers to evaluate performance on unseen data.

- **Merging Data:** We have combined Formula 1 records with Formula 2 (junior series) statistics using driver names as the key.
- **Handling Missing Values:** Any missing numeric data (like age or race starts) was filled using the median since it is less affected by outliers.
- **Encoding Categorical Data:** Text-based information such as debut teams and weather conditions was converted into numeric labels using *Label Encoding* so the model could process them.
- **Feature Scaling:** Numeric columns are standardized using *StandardScaler* to ensure that all features are contributing equally for the model training.

These preprocessing steps are needed to make our data uniform and ready for machine learning.

```
from sklearn.preprocessing import LabelEncoder, StandardScaler

X[categorical_columns] =
X[categorical_columns].apply(LabelEncoder().fit_transform)
X[numeric_columns] = StandardScaler().fit_transform(X[numeric_columns])
```

Building the Prediction Model:
We use a logistic regression model — a simple but powerful method which helps us to evaluate binary data such as whether the driver will score points or not.

Estimating a probability is how logistic regression operates:

$$P(\text{scored}) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \ldots)}}$$

Logistic regression works by estimating the probabilities, where coefficient (β) shows how much a feature like wins or debut age have an effect on the chance of scoring points.

We split the dataset into 75% for training and 25% for testing to ensure evaluation.The model learns how to recognize scoring patterns from the training data, and the testing data is then used to see how well it can make predictions on new drivers

```
X_train, X_test, y_train, y_test = train_test_split(X, y, stratify=y,
test_size=0.25)
model = LogisticRegression(max_iter=1000, class_weight='balanced')
model.fit(X_train, y_train)
```

Prediction Data (Rookies):
Contains information for drivers yet to debut in Formula 1, including only their pre-F1 statistics such as:

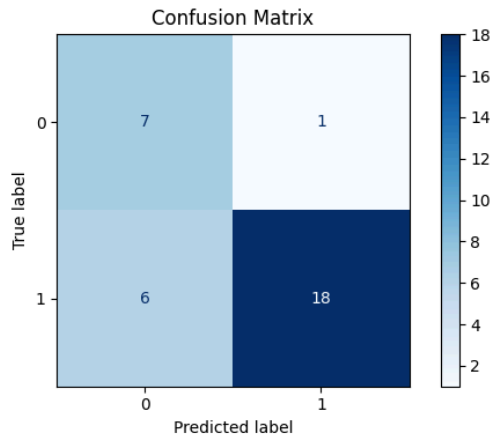
- F2 records (wins, podiums, crashes, championships)
- Age and debut team details

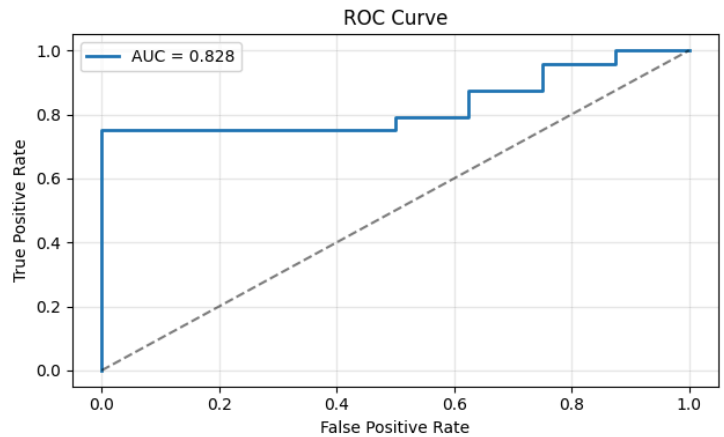
Once trained, we evaluated the model using key classification metrics:

- **Accuracy:** overall correctness of predictions.
- **Precision:** how many predicted scorers actually scored.
- **Recall:** how many actual scorers were correctly identified.

- **F1 Score:** balance between precision and recall.
- **ROC AUC:** how well the model separates point scoring drivers from non-scorers.

Together, these metrics show whether the model can make reliable predictions for rookie drivers.



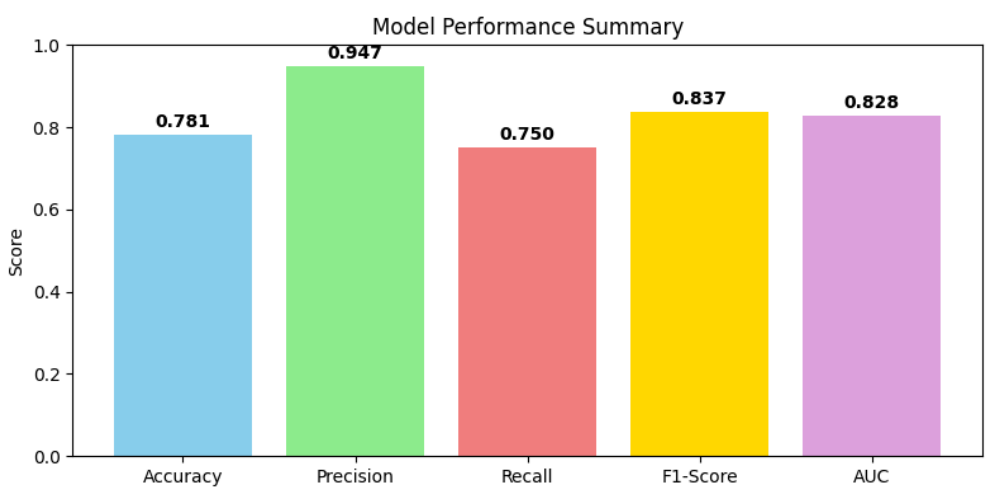


Logistic Regression provides coefficients that indicate how much each parameter contributes in scoring points.

By converting these into **odds ratios**, we can interpret them more intuitively:

- Features with odds > 1 increase the chances of scoring points.
- Features with odds < 1 reduce it.

In our model, features like **race wins, podium finishes**, and **number of race entries have** shown the strongest positive influence, while a **higher number of DNFs (Did Not Finish)** had a negative impact.



Although it's never easy to predict rookie success, data helps us reveal hidden trends underlying successful debuts. In the end, our method demonstrates how machine learning and motorsport interact, enabling us to comprehend what makes a driver exceptional from their first race weekend.

```
--- 2025 rookies predicted probability of scoring points ---
          driver_name  pred_prob_scored_points
0  Gabriel Bortoleto                 0.799724
1        Isack Hadjar                0.799724
2      Kimi Antonelli                0.834344
3     Oliver Bearman                 0.834344
```

Conclusion

We have tried to reveal the relationships and estimate team performance, dependability and risk in formula one sport using statistical modelling and machine learning. While our K-means clustering is identifying trends in the driver risk behavior, correlation and reliability allows us to relate how consistently propels team performance. Predicting rookie potential based on pre-F1 accomplishments is further made possible by logistic regression. When we take it as a whole, our outputs are a reflection of how data drivers analysis helps decipher racing dynamics and provides a fresh and quantitative viewpoint on what goes into essentially creating the perfect formula one team and champion.

Data Sources

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